REMOVING STRICTLY STANDARD ELEMENTS FROM A RESOLUTION ONE AT A TIME

THE STACKS PROJECT AUTHORS

Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Lifting a mod-pi^8 resolution to a solution. lemma-desingularize-lifting-apply (carded L26809). Performs the r = 1 step of the induction.
- Strictly standard elements survive base change. lemma-strictly-standard-base-change (carded L26796). Keeps hypothesis (2) true after quotienting by pi_1^8 in the r > 1 step.

Source reference. Original lemma 07F0: sources/stacks-smoothing/pinned-original-source.tex, source lines 2079–2079. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CC: sources/stacks-smoothing/pinned-original-source.tex, source lines 512–512. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07F7: sources/stacks-smoothing/pinned-original-source.tex, source lines 2296–2296. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $R \to A \to \Lambda \supset \mathfrak{q}$ be as in Situation 07F7. Let $r \geq 1$ and $\pi_1, \ldots, \pi_r \in R$ map to elements of $\mathfrak{q}$. Assume*

(1) *for $i = 1, \ldots, r$ we have*

$$Ann_{R/(\pi_1^8, \ldots, \pi_{i-1}^8)R}(\pi_i) = Ann_{R/(\pi_1^8, \ldots, \pi_{i-1}^8)R}(\pi_i^2)$$

and

$$Ann_{\Lambda/(\pi_1^8, \ldots, \pi_{i-1}^8)\Lambda}(\pi_i) = Ann_{\Lambda/(\pi_1^8, \ldots, \pi_{i-1}^8)\Lambda}(\pi_i^2)$$

(2) *for $i = 1, \ldots, r$ the element π_i maps to a strictly standard element in A over R.*

Then, if

$$R/(\pi_1^8, \ldots, \pi_r^8)R \to A/(\pi_1^8, \ldots, \pi_r^8)A \to \Lambda/(\pi_1^8, \ldots, \pi_r^8)\Lambda \supset \mathfrak{q}/(\pi_1^8, \ldots, \pi_r^8)\Lambda$$

can be resolved, so can $R \to A \to \Lambda \supset \mathfrak{q}$.

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Proof. We are going to prove this by induction on r.

The case $r = 1$. Here the assumption is that there exists a factorization $A/\pi_1^8 \to \bar{C} \to \Lambda/\pi_1^8$ which resolves the situation modulo π_1^8. Conditions (1) and (2) are the assumptions needed to apply Lemma 07F0. Thus we can "lift" the resolution $\bar{C}$ to a resolution of $R \to A \to \Lambda \supset \mathfrak{q}$.

The case $r > 1$. In this case we apply the induction hypothesis for $r - 1$ to the situation $R/\pi_1^8 \to A/\pi_1^8 \to \Lambda/\pi_1^8 \supset \mathfrak{q}/\pi_1^8\Lambda$. Note that property (2) is preserved by Lemma 07CC. $\square$